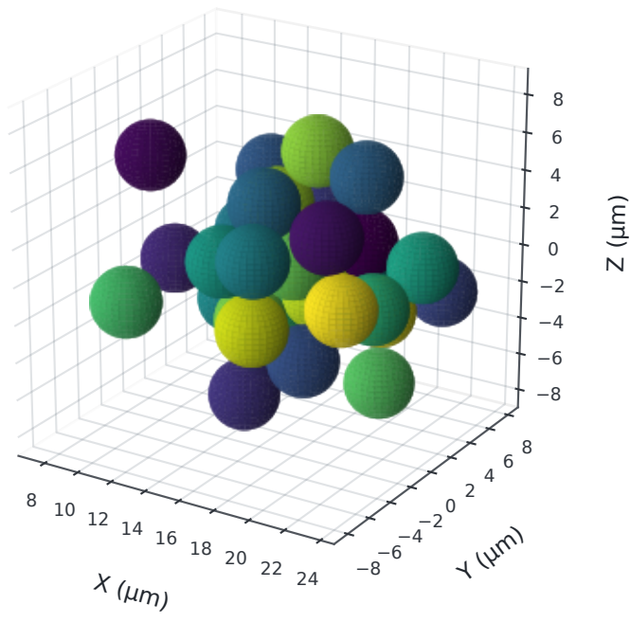
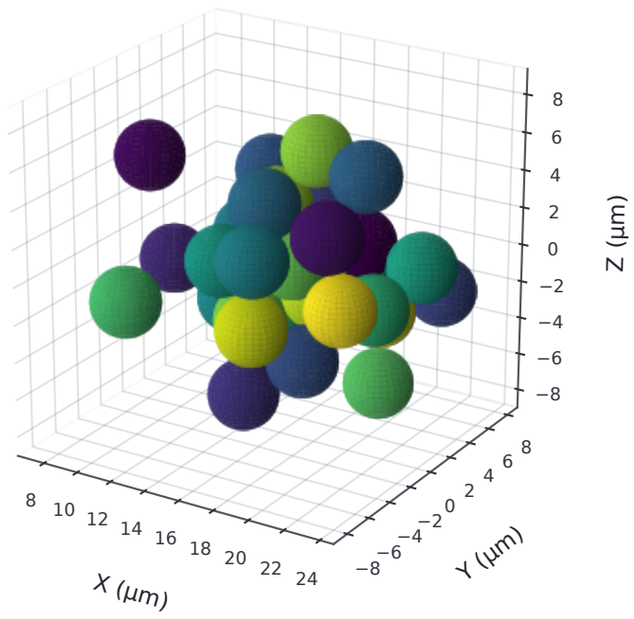
F10F | Fixed-ID geometry remains numerically sensitive

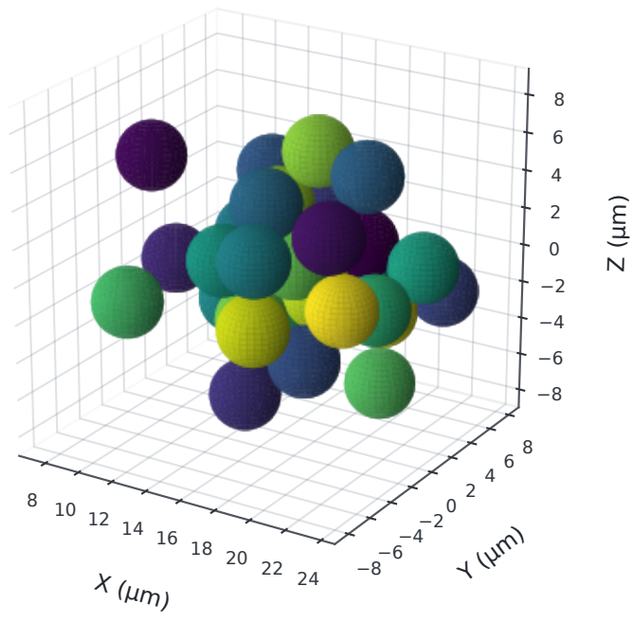
**a B0 | accepted 1.700 ms**



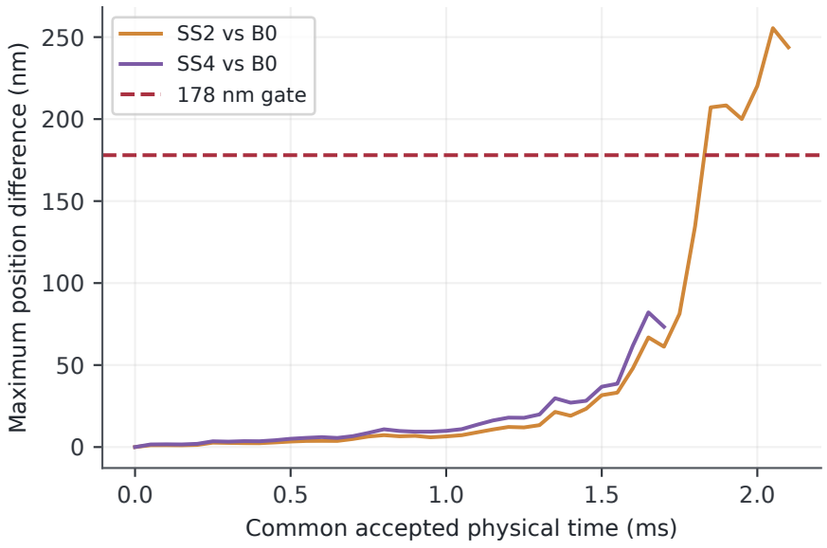

**b SS2 | accepted 1.700 ms**



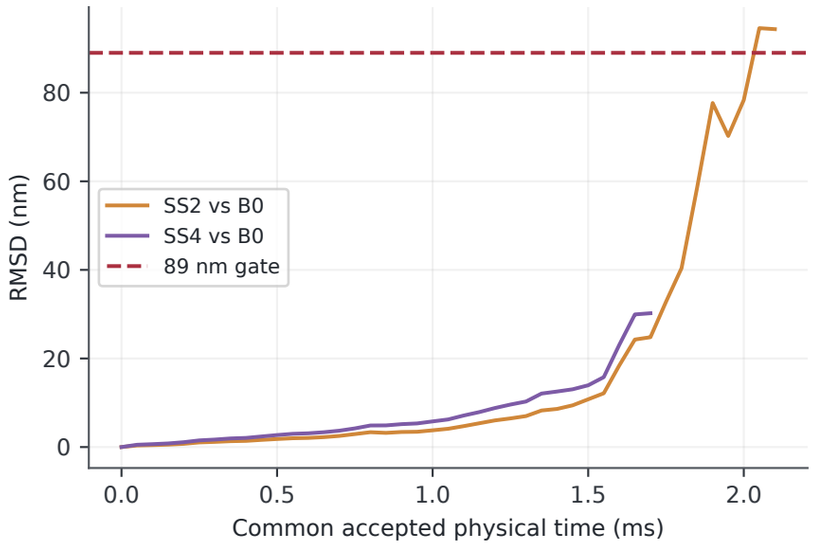

**c SS4 | accepted 1.700 ms**



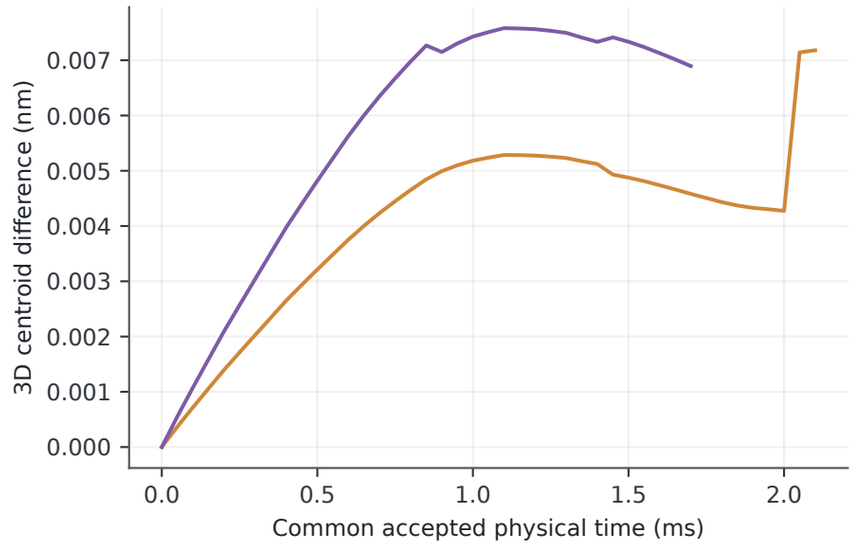

**d Fixed-ID position sensitivity**



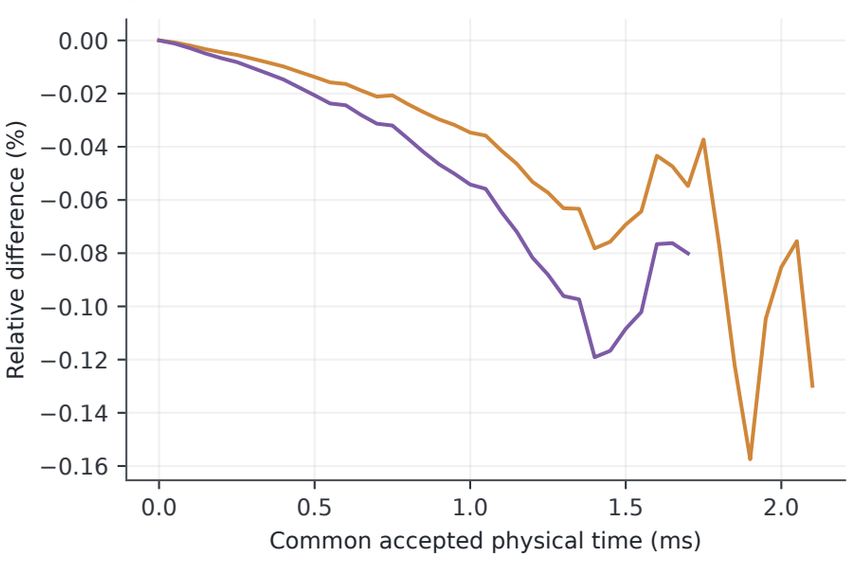

**e Kabsch-aligned shape sensitivity**



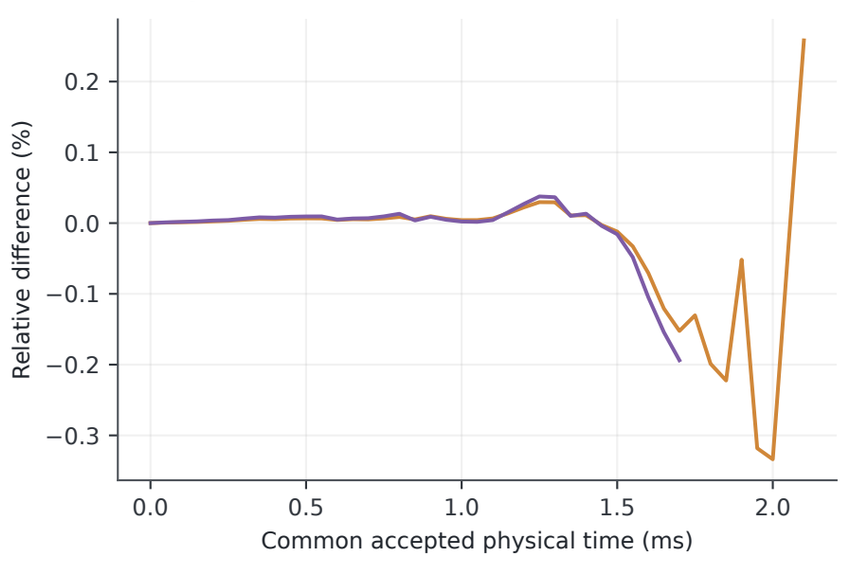

**f Centroid sensitivity**



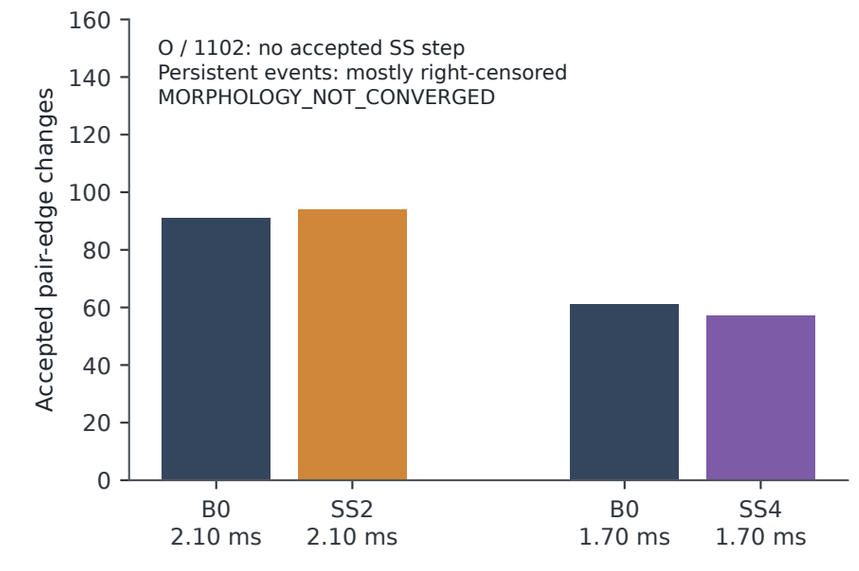

**g Radius of gyration**




**h Aspect ratio**




**i Edge events and evidence limits**




Preparation / seed 1104. True radius = 1.78 μm; identical origin, 3D limits and camera. Fixed ID colors: viridis, 0 → 31. No rejected state is drawn as accepted.